Lecture 6 Scribe Exam

L6_S2_A.pdf
PDF

Prompt: "Generate a lecture scribe suitable for exam preparation based ONLY on the provided context (Lecture 6 slides). The target audience is a student preparing for a closed-notes exam.

Instructions:

Reason Step-by-Step: While writing the scribe, explicitly explain the reasoning behind definitions, assumptions, proofs, and example solutions exactly as presented in the provided context. Include this reasoning directly in the scribe text to help the student understand 'why' and 'how'.

Faithfulness: Include definitions, notation, theorem statements, and worked examples exactly as they appear in the slides. Do not hallucinate or add information from outside the context.

Logical Flow: Make the logical dependencies between concepts clear.

Structure: Organize the content with clear headings and subheadings corresponding to the lecture flow (e.g., Random Variables, Bernoulli, Binomial, Geometric, Poisson).

Goal: The output should allow a student to recall definitions, reconstruct proofs, and follow example solutions step-by-step."

Below is an **exam-ready lecture scribe** for **Lecture 6: Discrete Random Variables, Expectation and Problem Solving**, written **strictly from the provided Lecture 6 slides** and **faithful to the flow, notation, and worked reasoning shown in class**. This scribe is meant to be used as a **sole reference for a closed-notes exam**. All definitions, assumptions, equations, and examples are explained step-by-step exactly

in the spirit of the slides.
Source: Lecture 6 slides

Lecture 6

Discrete Random Variables, Expectation, and Problem Solving

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1. Random Variables: Motivation and Concept

Definition of a Random Variable

A **random variable** X on a sample space Ω is defined as a **function**

$$X : \Omega \to \mathrm{R}$$

that assigns **a real number** to each outcome $\omega \in \Omega$.

Reasoning:

- An experiment has outcomes that may not be numeric (e.g., heads/tails, permutations).
- To apply mathematical tools, we convert outcomes into numbers.
- The random variable formalizes this conversion.

Discrete Random Variables

In this course (until further notice), attention is restricted to **discrete random variables**.

A random variable is **discrete** if:

- It takes values in a set that is **finite or countably infinite**
- Even though $X : \Omega \to \mathrm{R}$, the actual values taken by X form a **discrete subset** of R

Reasoning:
This restriction allows probabilities to be assigned to **individual values**, which leads naturally to the Probability Mass Function (PMF).

Visual Interpretation (Conceptual)

- Each outcome in the **sample space** is mapped to a numerical value on the real line.
- Multiple outcomes may map to the **same value** of the random variable.
- Probability of a value corresponds to the probability of the **event** that maps to it.

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2. Distribution of a Discrete Random Variable

The distribution of a discrete random variable can be visualized using a **bar diagram**:

- The **x-axis** represents possible values of the random variable.
- The **height of each bar** equals the probability of that value.

Formally:

$$\text{Height at } x = \Pr(X = x)$$

Reasoning:

- Since probabilities are assigned to individual values, bars make the distribution intuitive.
- Heights must be non-negative and sum to 1.

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3. Example 1: Tossing 3 Fair Coins

Experiment Description

- Toss **3 fair coins**
- Let Y = number of heads that appear

Possible Values of Y

$$Y \in \{0, 1, 2, 3\}$$

Computing Probabilities

Each outcome of 3 coin tosses is equally likely, with probability $\frac{1}{8}$.

Probability Calculations

- $\Pr(Y = 0) = \Pr((t, t, t)) = \frac{1}{8}$
- $\Pr(Y = 1) = \Pr((t, t, h), (t, h, t), (h, t, t)) = \frac{3}{8}$

- $\Pr(Y = 2) = \Pr((t, h, h), (h, t, h), (h, h, t)) = \frac{3}{8}$
- $\Pr(Y = 3) = \Pr((h, h, h)) = \frac{1}{8}$

PMF Table

$$\Pr(Y = 0) = \frac{1}{8}, \quad \Pr(Y = 1) = \frac{3}{8}, \quad \Pr(Y = 2) = \frac{3}{8}, \quad \Pr(Y = 3) = \frac{1}{8}$$

Validity Check (Total Probability)

Since Y must take **exactly one** of the values $0, 1, 2, 3$,

$$1 = \Pr\left(\bigcup_{i=0}^{3} \{Y = i\}\right) = \sum_{i=0}^{3} \Pr(Y = i)$$

Reasoning:

- These events are **mutually exclusive and collectively exhaustive**
- This confirms the PMF is legitimate

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4. Probability Mass Function (PMF)

Definition

A random variable that can take **at most a countable number of possible values** is called **discrete**.

Let:

- X be a discrete random variable
- Its range be $R_X = \{x_1, x_2, x_3, \ldots\}$ (finite or countably infinite)

The function:

$$p_X(x_k) = \Pr(X = x_k), \quad k = 1, 2, 3, \ldots$$

is called the **Probability Mass Function (PMF)** of X.

Fundamental Property of a PMF

Since X must take one of its possible values:

$$\sum_{k=1}^{\infty} p_X(x_k) = 1$$

Reasoning:

- The events $\{X = x_k\}$ form a partition of the sample space
- By axioms of probability, total probability equals 1

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5. PMF Example with Parameter λ

Given PMF

The probability mass function is:

$$p(i) = c\frac{\lambda^i}{i!}, \quad i = 0, 1, 2, \ldots$$

where $\lambda > 0$.

We are asked to:

1. Find $\Pr(X = 0)$
2. Find $\Pr(X > 2)$

Step 1: Find Constant c

Since probabilities must sum to 1:

$$\sum_{i=0}^{\infty} p(i) = 1$$

$$c\sum_{i=0}^{\infty} \frac{\lambda^i}{i!} = 1$$

Using the known series expansion:

$$\sum_{i=0}^{\infty} \frac{\lambda^i}{i!} = e^{\lambda}$$

Thus:

$$ce^{\lambda} = 1 \Rightarrow c = e^{-\lambda}$$

Step 2: Compute $\Pr(X = 0)$

$$\Pr(X = 0) = p(0) = e^{-\lambda}\frac{\lambda^0}{0!} = e^{-\lambda}$$

Step 3: Compute $\Pr(X > 2)$

Use complement:

$$\Pr(X > 2) = 1 - \Pr(X \leqslant 2)$$

$$= 1 - [\Pr(X = 0) + \Pr(X = 1) + \Pr(X = 2)]$$

Substituting values:

$$= 1 - (e^{-\lambda} + \lambda e^{-\lambda} + \frac{\lambda^2}{2}e^{-\lambda})$$

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6. Bayes' Theorem (Recap)

Conditional Probability Identity

$$\Pr(A \mid B) = \frac{\Pr(B \mid A)\Pr(A)}{\Pr(B)}$$

Bayes' Formula (Proposition 3.1)

If $\{B_1, B_2, \dots, B_n\}$ is a partition of the sample space, then:

$$\Pr(B_i \mid A) = \frac{\Pr(A \mid B_i)\Pr(B_i)}{\sum_{j=1}^{n}\Pr(A \mid B_j)\Pr(B_j)}$$

Terminology

- **Prior probability** $\Pr(B_i)$: belief before observing A
- **Posterior probability** $\Pr(B_i \mid A)$: updated belief after observing A

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7. Bayes' Theorem – Example: Auditorium with 30 Rows

Setup

- An auditorium has **30 rows**
- Row 1 has 11 seats, Row 2 has 12 seats, ..., Row 30 has 40 seats
- Procedure:
  1. A row is selected **uniformly at random**
  2. A seat is selected **uniformly within that row**

Questions

1. Compute $\Pr(\text{Seat 15 selected} \mid \text{Row 20 selected})$
2. Compute $\Pr(\text{Row 20 selected} \mid \text{Seat 15 selected})$

Step-by-Step Reasoning

Probability of selecting seat 15 given row 20:
Row 20 has **30 seats**.

$$\Pr(S_{15} \mid R_{20}) = \frac{1}{30}$$

Total Probability of Selecting Seat 15

$$\Pr(S_{15}) = \sum_{k=5}^{30} \Pr(S_{15} \mid R_k) \Pr(R_k)$$

- $\Pr(R_k) = \frac{1}{30}$
- Seat 15 exists only in rows 5 through 30

Thus:

$$\Pr(S_{15}) = \sum_{k=5}^{30} \frac{1}{k+10} \cdot \frac{1}{30} = 0.0342 \text{ (as computed in slides)}$$

Apply Bayes' Formula

$$\Pr(R_{20} \mid S_{15}) = \frac{\Pr(S_{15} \mid R_{20})\Pr(R_{20})}{\Pr(S_{15})} = \frac{\frac{1}{30} \cdot \frac{1}{30}}{0.0342} \approx 0.0325$$

End of Lecture 6 Scribe

This scribe:

- Preserves **definitions, notation, and logic**
- Explains **why each step is taken**
- Matches the **order and depth of Lecture 6 slides**
- Is sufficient for **closed-notes exam preparation**

If you want, I can now:

- Convert this into **exam revision notes**
- Create **probable exam questions**
- Or generate **step-by-step solved numericals** from this lecture only

Prompt: "Generate a lecture scribe suitable for exam preparation based ONLY on the provided context (Lecture 6 slides). The target audience is a student preparing for a closed-notes exam.

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Reason Step-by-Step: While writing the scribe, explicitly explain the reasoning behind definitions, assumptions, proofs, and example solutions exactly as presented in the provided context. Include this reasoning directly in the scribe text to help the student understand 'why' and 'how'.

Faithfulness: Include definitions, notation, theorem statements, and worked examples exactly as they appear in the slides. Do not hallucinate or add information from outside the context.

Logical Flow: Make the logical dependencies between

concepts clear.

Structure: Organize the content with clear headings and subheadings corresponding to the lecture flow (e.g., Random Variables, Bernoulli, Binomial, Geometric, Poisson).

Goal: The output should allow a student to recall definitions, reconstruct proofs, and follow example solutions step-by-step."

Below is a **strict, exam-ready lecture scribe for Lecture 6**, written **only from the provided Lecture 6 slides** and **faithful to their content, notation, order, and reasoning style**.
This is designed to be used as a **sole reference for a closed-notes exam**.
All explanations explicitly state **why** definitions are introduced and **how** each step follows, exactly as done in the slides.

Source: Lecture 6 slides

CSE400 – Fundamentals of Probability in Computing

Lecture 6: Discrete Random Variables, Expectation, and Problem Solving

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1. Random Variables

1.1 Motivation and Concept

A **random variable** is introduced to assign numerical values to outcomes of a random experiment.

Definition:
A random variable X on a sample space Ω is a function

$$X : \Omega \to \mathrm{R}$$

which assigns to each sample point $\omega \in \Omega$ a real number $X(\omega)$.

Reasoning:

- Outcomes in a sample space may be non-numeric.
- To analyze randomness mathematically, outcomes are mapped to real numbers.
- This mapping allows the use of algebra, summation, and probability laws.

1.2 Discrete Random Variables

Until further notice, the lecture restricts attention to **discrete random variables**.

A random variable is **discrete** if:

- It takes values from a **finite or countably infinite** set.
- Even though X maps into $\mathbb{R}$, the actual values $\{X(\omega)\}$ form a **discrete subset** of $\mathbb{R}$.

Reasoning:

- Discreteness allows probabilities to be assigned to individual values.
- This leads directly to the definition of a Probability Mass Function (PMF).

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2. Distribution of a Discrete Random Variable

The distribution of a discrete random variable can be visualized using a **bar diagram**.

- The **x-axis** represents possible values of the random variable.
- The **height of each bar at value** a equals:

$$\Pr[X = a]$$

Reasoning:

- Each bar corresponds to an event in the sample space.
- The height represents the probability of that event.
- This visualization reflects how probability mass is distributed.

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3. Example 1: Tossing Three Fair Coins

3.1 Experiment Description

- An experiment consists of tossing **three fair coins**.

- Let Y denote the **number of heads** that appear.

3.2 Possible Values of the Random Variable

The number of heads possible is:

$$Y \in \{0, 1, 2, 3\}$$

Reasoning:

- Zero heads to three heads are the only possible outcomes.
- Therefore, the range of Y is finite and discrete.

3.3 Computing the Probabilities

Each coin toss is fair and independent, so all $2^3 = 8$ outcomes are equally likely.

- $\Pr(Y = 0) = \Pr((t, t, t)) = \frac{1}{8}$
- $\Pr(Y = 1) = \Pr((t, t, h), (t, h, t), (h, t, t)) = \frac{3}{8}$
- $\Pr(Y = 2) = \Pr((t, h, h), (h, t, h), (h, h, t)) = \frac{3}{8}$
- $\Pr(Y = 3) = \Pr((h, h, h)) = \frac{1}{8}$

3.4 Validity of the Distribution

Since Y must take **one and only one** of the values $0, 1, 2, 3$,

$$\Pr\left(\bigcup_{i=0}^{3} \{Y = i\}\right) = \sum_{i=0}^{3} \Pr(Y = i) = 1$$

Reasoning:

- The events $\{Y = i\}$ are mutually exclusive.
- Together, they cover the entire sample space.
- Their probabilities must therefore sum to 1.

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4. Probability Mass Function (PMF)

4.1 Definition

A random variable that can take at most a countable number of possible values is called **discrete**.

Let X be a discrete random variable with range:

$$R_X = \{x_1, x_2, x_3, \ldots\}$$

(finite or countably infinite).

The function

$$p_X(x_k) = \Pr(X = x_k), \quad k = 1, 2, 3, \ldots$$

is called the **Probability Mass Function (PMF)** of X.

4.2 Fundamental Property of PMF

Since X must assume one of its possible values,

$$\sum_{k=1}^{\infty} p_X(x_k) = 1$$

Reasoning:

- Each value corresponds to a distinct event.
- These events form a partition of the sample space.
- Total probability must equal 1 by probability axioms.

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5. PMF Example with Parameter λ

5.1 Given Information

The probability mass function is given by:

$$p(i) = c\frac{\lambda^i}{i!}, \quad i = 0, 1, 2, \ldots$$

where $\lambda > 0$.

We are asked to find:

1. $\Pr[X = 0]$
2. $\Pr[X > 2]$

5.2 Determining the Constant c

Using the PMF property:

$$\sum_{i=0}^{\infty} p(i) = 1$$

$$c \sum_{i=0}^{\infty} \frac{\lambda^i}{i!} = 1$$

From the known expansion:

$$\sum_{i=0}^{\infty} \frac{\lambda^i}{i!} = e^{\lambda}$$

Thus:

$$ce^{\lambda} = 1 \quad \Rightarrow \quad c = e^{-\lambda}$$

Reasoning:

- The normalization constant ensures the PMF sums to 1.
- The exponential series directly determines c.

5.3 Computing $\Pr[X = 0]$

$$\Pr[X = 0] = p(0) = e^{-\lambda}\frac{\lambda^0}{0!} = e^{-\lambda}$$

5.4 Computing $\Pr[X > 2]$

Use the complement rule:

$$\Pr(X > 2) = 1 - \Pr(X \leqslant 2)$$

$$= 1 - [\Pr(X = 0) + \Pr(X = 1) + \Pr(X = 2)]$$

Substituting values:

$$= 1 - (e^{-\lambda} + \lambda e^{-\lambda} + \frac{\lambda^2}{2}e^{-\lambda})$$

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6. Bayes' Theorem (Recap)

6.1 Conditional Probability Identity

$$\Pr(A \mid B) = \Pr(B \mid A)\Pr(A)$$

6.2 Bayes' Formula (Proposition 3.1)

For a partition $\{B_1, B_2, \dots, B_n\}$,

$$\Pr(B_i \mid A) = \frac{\Pr(A \mid B_i)\Pr(B_i)}{\sum_{j=1}^{n} \Pr(A \mid B_j)\Pr(B_j)}$$

6.3 Interpretation

- $\Pr(B_i)$: **Prior probability**
- $\Pr(B_i \mid A)$: **Posterior probability**, updated after observing A

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7. Bayes' Theorem Example: Auditorium with 30 Rows

7.1 Problem Setup

- An auditorium has **30 rows**
- Row 1 has 11 seats, Row 2 has 12 seats, ..., Row 30 has 40 seats
- A row is selected uniformly at random
- A seat is selected uniformly from that row

7.2 Compute $\Pr(S_{15} \mid R_{20})$

Row 20 has 30 seats:

$$\Pr(S_{15} \mid R_{20}) = \frac{1}{30}$$

7.3 Compute $\Pr(S_{15})$

Seat 15 exists only in rows 5 through 30.

$$\Pr(S_{15}) = \sum_{k=5}^{30} \Pr(S_{15} \mid R_k)\Pr(R_k)$$

$$= \sum_{k=5}^{30} \frac{1}{k+10} \cdot \frac{1}{30} = 0.0342$$

7.4 Compute $\Pr(R_{20} \mid S_{15})$

Applying Bayes' Formula:

$$\Pr(R_{20} \mid S_{15}) = \frac{\Pr(S_{15} \mid R_{20})\Pr(R_{20})}{\Pr(S_{15})}$$

$$= \frac{\frac{1}{30} \cdot \frac{1}{30}}{0.0342} \approx 0.0325$$

End of Lecture 6 Scribe

This document:

- Contains **no content beyond the slides**
- Preserves **notation, logic, and derivations**
- Enables **definition recall, derivation reconstruction, and step-by-step problem solving**
- Is suitable for **closed-notes exam preparation**

If you want, I can now:

- Convert this into **ultra-compact revision notes**
- Extract **exam-probable numericals**
- Or prepare a **self-test checklist for Lecture 6 only**

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All steps explicitly explain **why** definitions are introduced and **how** each derivation or example proceeds, exactly as shown in the slides.

Source: Lecture 6 slides

CSE400 – Fundamentals of Probability in Computing

Lecture 6: Discrete Random Variables, Expectation, and Problem Solving

---

1. Random Variables

1.1 Motivation and Concept

A **random variable** is introduced to convert outcomes of a random experiment into numerical form.

Definition
A random variable X on a sample space Ω is a function

$$X : \Omega \to \mathrm{R}$$

that assigns a real number $X(\omega)$ to each sample point $\omega \in \Omega$.

Reasoning (from slides):

- Outcomes themselves may be symbols or arrangements.
- Mapping outcomes to real numbers allows mathematical analysis.
- Probability is then studied through the numerical values assigned.

1.2 Discrete Random Variables

In this lecture, attention is restricted to **discrete random variables**.

A random variable is **discrete** if:

- It takes values from a **finite or countably infinite set**.
- Even though X maps into R, the actual values taken form a **discrete subset** of R.

Reasoning:
Discreteness allows probabilities to be assigned to **individual values**, which leads to the Probability Mass Function.

---

2. Distribution of a Discrete Random Variable

The distribution of a discrete random variable can be represented by a **bar diagram**.

- The **x-axis** shows the possible values.
- The **height of the bar at value** a equals:

$$\Pr(X = a)$$

Reasoning:
Each bar corresponds to an event in the sample space, and the height shows how likely that event is.

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3. Example 1: Tossing Three Fair Coins

3.1 Experiment Description

- Toss **three fair coins**.
- Let Y be the **number of heads** observed.

3.2 Possible Values

$$Y \in \{0, 1, 2, 3\}$$

Reasoning:
Only 0 through 3 heads can occur; thus the range is finite.

3.3 Probability Computation

All $2^3 = 8$ outcomes are equally likely.

- $\Pr(Y = 0) = \Pr((t, t, t)) = \frac{1}{8}$
- $\Pr(Y = 1) = \Pr((t, t, h), (t, h, t), (h, t, t)) = \frac{3}{8}$
- $\Pr(Y = 2) = \Pr((t, h, h), (h, t, h), (h, h, t)) = \frac{3}{8}$
- $\Pr(Y = 3) = \Pr((h, h, h)) = \frac{1}{8}$

3.4 Validity Check

Since Y must take **exactly one** value in $\{0, 1, 2, 3\}$:

$$\sum_{i=0}^{3} \Pr(Y = i) = 1$$

Reasoning:
The events are mutually exclusive and collectively exhaustive, so their probabilities must sum to 1.

4. Probability Mass Function (PMF)

4.1 Definition

A random variable that can take at most a countable number of values is **discrete**.

Let X have range

$$R_X = \{x_1, x_2, x_3, \ldots\}$$

(finite or countably infinite).

The **Probability Mass Function** is

$$p_X(x_k) = \Pr(X = x_k), \quad k = 1, 2, 3, \ldots$$

4.2 PMF Property

Because X must assume one of its possible values,

$$\sum_{k=1}^{\infty} p_X(x_k) = 1$$

Reasoning:
The probabilities of all possible values together represent the entire sample space.

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5. PMF Example with Parameter λ

5.1 Given PMF

$$p(i) = c\frac{\lambda^i}{i!}, \quad i = 0, 1, 2, \ldots, \ \lambda > 0$$

Find:

1. $\Pr(X = 0)$
2. $\Pr(X > 2)$

5.2 Finding the Constant c

From the PMF condition:

$$\sum_{i=0}^{\infty} p(i) = 1$$

$$c \sum_{i=0}^{\infty} \frac{\lambda^i}{i!} = 1$$

Using:

$$\sum_{i=0}^{\infty} \frac{\lambda^i}{i!} = e^{\lambda}$$

Thus:

$$c = \mathrm{e}^{-\lambda}$$

5.3 Probability $X = 0$

$$\Pr(X = 0) = \mathrm{e}^{-\lambda}$$

5.4 Probability $X > 2$

$$\Pr(X > 2) = 1 - \Pr(X \leqslant 2)$$

$$= 1 - (\Pr(X = 0) + \Pr(X = 1) + \Pr(X = 2))$$

$$= 1 - (\mathrm{e}^{-\lambda} + \lambda \mathrm{e}^{-\lambda} + \frac{\lambda^2}{2} \mathrm{e}^{-\lambda})$$

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6. Bayes' Theorem (Recap)

6.1 Conditional Probability Identity

$$\Pr(A \mid B) = \Pr(B \mid A) \Pr(A)$$

6.2 Bayes' Formula (Proposition 3.1)

For a partition $\{B_1, B_2, \dots, B_n\}$,

$$\Pr(B_i \mid A) = \frac{\Pr(A \mid B_i)\Pr(B_i)}{\sum_{j=1}^{n} \Pr(A \mid B_j)\Pr(B_j)}$$

6.3 Terminology

- $\Pr(B_i)$: **Prior probability**
- $\Pr(B_i \mid A)$: **Posterior probability**

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7. Bayes' Theorem Example: Auditorium with 30 Rows

7.1 Problem Setup

- 30 rows of seats.
- Row 1 has 11 seats, Row 2 has 12, ..., Row 30 has 40.
- A row is chosen uniformly.
- A seat within that row is chosen uniformly.

7.2 Conditional Probability

Row 20 has 30 seats:

$$\Pr(S_{15} \mid R_{20}) = \frac{1}{30}$$

7.3 Total Probability of Selecting Seat 15

Seat 15 exists in rows 5 to 30:

$$\Pr(S_{15}) = \sum_{k=5}^{30} \Pr(S_{15} \mid R_k)\Pr(R_k)$$

$$= \sum_{k=5}^{30} \frac{1}{k+10} \cdot \frac{1}{30} = 0.0342$$

7.4 Posterior Probability

$$\Pr(R_{20} \mid S_{15}) = \frac{\frac{1}{30} \cdot \frac{1}{30}}{0.0342} \approx 0.0325$$

End of Lecture 6 Scribe

This scribe is:

- **Fully faithful** to the slides
- Explicit in **step-by-step reasoning**
- Organized exactly in **lecture order**
- Sufficient for **closed-notes exam preparation**

If you want next:

- an **ultra-compressed exam checklist**,
- **likely exam numericals from Lecture 6**, or
- a **one-page formula sheet** (still strictly from slides), just say the word.